PIPELINE

PHASE 1 — DOCUMENT STRATEGY

What to do

- Choose ONE legal domain (BNS 2023)
- Define metadata

Output

```
{
  "act": "BNS 2023",
  "version_date": "2023-12-25",
  "jurisdiction": "India"
}
```

Why

Prevents ambiguity + future updates

PHASE 2 — INGESTION (CLEAN TEXT)

What to do

- Extract PDF → text
- Remove noise (headers, page numbers)
- Keep structure

Output

raw_cleaned.txt

Why

Garbage in = garbage out
This step ensures clean foundation

PHASE 3 — STRUCTURE PARSING

What to do

Convert text → structured legal units

Output: sections.json

```
{
    "doc_id": "",
```

```
    "act": "",
    "chapter_number":"",
    "chapter_title":"",
    "section_number": 1,
    "section_tile": 1,
    "order": 1,
    "chapter_order": 1,
    "clauses": [
      {
        "clause_id": "",
        "clause_number": "",
        "text": ""
      },
      {},
      {}
    ],
    "explanation":[
      {
        "text": ""
      },
      {}
    ],
    "illustration":[
      {
        "text": ""
      },
      {}
    ],
    "full_text": ""
  }
```


Why

Law becomes **machine-readable + queryable**

PHASE 4 — ENRICHMENT

What to do

Add intelligence to sections

Output: enriched_sections.json


```
{
    "doc_id": "",
    "section_number": 1,
    "section_type": "",
    "keywords": ["", ""],
    "legal_terms": ["", ""],
    "importance_score": 1,
    "cross_reference":[
      {
        "target_section": 1,
        "target_doc_id": "",
        "type": "",
        "context": ""
      },
      {},
      {}
    ]
  }
```


Why

Helps retrieval + reasoning later

PHASE 5 — GRAPH BUILDING

What to do

Extract relationships between sections

Output: graph.json

```
{
  "BNS_2023_S10": ["BNS_2023_S2"]
}
```

Why

Enables **multi-hop reasoning (law is interconnected)**

PHASE 6 — CHUNKING

What to do

Break sections → chunks (small units)

Output: chunks.json

```
{
    "chunk_id": "",
    "section_id": 1,
    "chapter_number":"",
    "section_number": 1,
    "order": 1,
    "chapter_order": 1,
    "type": "",
    "text": "",
    "token_length": "",
    "keywords": ["", ""],
    "importance": 1,
    "embedding_ready_text": ""
  }
```

Why

Retrieval works on **small precise units**

PHASE 7 — EMBEDDINGS + INDEXING

What to do

- Convert chunks → vectors
- Store in FAISS
- Build BM25 index

Output

Vector DB + keyword index

Why

Enables **semantic + keyword search**

PHASE 8 — QUERY ANALYSIS (LLM)

What to do

Understand user query

Output

```
{
  "intent_type": "definition",
  "target": "animal"
}
```

Why

Guides entire pipeline (retrieval → reasoning)

PHASE 9 — HYBRID RETRIEVAL

What to do

Search using:

- Vector search (semantic)
- BM25 (keywords)

Output

```
[
  {"chunk_id": "BNS_S2_C2", "score": 0.92}
]
```

Why

Combines **meaning + exact match**

PHASE 10 — RE-RANKING (LLM)

What to do

Filter best chunks using LLM

Output

[
 {"chunk_id": "BNS_S2_C2", "score": 0.98}
]

Why

Removes noise → keeps only relevant law

PHASE 11 — MULTI-STAGE REASONING (LLM)

🔹 **STEP 11.1 — Confirm intent**

Validate query understanding

🔹 **STEP 11.2 — Select evidence**

Choose best chunks

🔹 **STEP 11.3 — Generate answer**

{
 "answer": "...",
 "citations": ["2"]
}

Uses ONLY law text

🔹 **STEP 11.4 — RISK ANALYSIS**

Checks:

- retrieval weak?
- chunk conflict?
- missing dependencies (graph)?
- reasoning valid?

Output

{
 "answer": "...",
 "risk_level": "low",
 "reasoning_score": 0.95
}

Why

Ensures **correct + explainable reasoning**

PHASE 12 — CONFIDENCE SCORING

What to do

Compute reliability numerically

Formula

```
confidence =
  0.4 * retrieval +
  0.3 * reranker +
  0.2 * citation +
  0.1 * reasoning
```

Output

```
{
  "confidence": 0.94
}
```

Why

System knows **how sure it is**

PHASE 13 — SAFETY & VALIDATION

What to do

Final strict checks

Checks:

- fake citations?
- no data retrieved?
- low confidence?
- hallucination?

Output (FINAL)

```
{
  "answer": "...",
  "citations": ["Section 2"],
  "confidence": 0.94,
  "risk_level": "low"
}
```

Why

Prevents **wrong or unsafe outputs**

FINAL SYSTEM FLOW (MENTAL MODEL)

Phase 1–6 → Build legal knowledge base
Phase 7 → Make it searchable
Phase 8–10 → Find best legal provisions
Phase 11 → Think like a lawyer
Phase 12 → Measure certainty
Phase 13 → Enforce correctness

INSIGHT

Chunks → find answers
Graph → find connections
LLM → reason
Math → confidence
Rules → safety